🔑 **Answer Key — Teacher Copy · Lesson 7-6**

MSTAR PRACTICE WORKSHEET

Lesson 7-6: Determine Distance on the Coordinate Plane

Grade 6 Mathematics · Standard 6.NOS.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the distance between (-4, 3) and (2, 3)?

A. 2 units

B. 4 units

C. 6 units

D. 8 units

Both points have y = 3, so this is a horizontal distance. |2 - (-4)| = |2 + 4| = 6 units.

Question 2 SELECTED RESPONSE — CORRECT: C

What is the distance between (5, -2) and (5, 4)?

A. 2 units

B. 4 units

C. 6 units

D. 10 units

Both points have x = 5, so this is a vertical distance. |4 - (-2)| = |4 + 2| = 6 units.

Question 3 SELECTED RESPONSE — CORRECT: C

A rectangle has vertices at (1, 1), (5, 1), (5, 4), (1, 4). What is its perimeter?

A 8 units

B 12 units

C 14 units

D 20 units

Width = 5 - 1 = 4. Height = 4 - 1 = 3. Perimeter = 2(4 + 3) = 14.

Question 4 SELECTED RESPONSE — CORRECT: D

Two points are at (-3, -7) and (-3, 5). What is the distance between them?

A 2 units

B 7 units

C 8 units

D 12 units

Same x = -3. Distance = |5 - (-7)| = |5 + 7| = 12 units.

Question 5 SELECTED RESPONSE — CORRECT: A

What is the distance between (2, 3) and (2, 8)?

A 5 units

B 5.5 units

C 6 units

D 11 units

The x-coordinates match, so distance = |8 - 3| = 5 units.

Question 6 SELECTED RESPONSE — CORRECT: C

What is the distance between (-4, 1) and (3, 1)?

A -7 units

B 1 unit

C 7 units

D 12 units

The y-coordinates match, so distance = |3 - (-4)| = 7 units.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A chest sits at (–3, 2) and a cave at (4, 2). How far apart are they?

A 1 unit

B 2 units

C 5 units

D 7 units

Both points share the same y-value, so the distance is horizontal: |–3 – 4| = 7 units.

- **A:** That subtracts the digits without regard to sign. Count from negative 3 up to zero, then on to 4.
- **B:** That is the shared y-value, not a distance. Measure along x.
- **C:** Both points sit at y = 2, so the gap is purely horizontal. Count along x from negative 3 to 4.

Part B — correct answer: A

Why can this distance be found by subtraction alone?

A The two points share the same y-coordinate, so the segment between them is horizontal.

B Subtraction works for any two points on the plane.

C Both coordinates are whole numbers.

D One of the points is in Quadrant II.

When two points share a coordinate, the segment runs straight along a grid line and its length is the difference of the coordinates that DO change. A slanted segment needs more than subtraction.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Subtracting a Negative

> Priya found the distance between (−3, 2) and (4, 2). She wrote: 'Just subtract: 4 − 3 = 1, so they are 1 unit apart.'

Explain Priya's misconception and give the correct distance.

Model answer: |4 − (−3)| = |4 + 3| = 7 units. Priya treated −3 as 3. Counting on the number line from −3 to 0 is 3 units and 0 to 4 is 4 more, confirming 7.

2 points	Student identifies that Priya dropped the negative sign on −3, sets up |4 − (−3)| = |4 + 3| = 7, and confirms by counting grid units from −3 to 4.
1 point	Student gives 7 units without explaining the dropped negative sign.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A rectangle has vertices at (-4, -2), (3, -2), (3, 5), and (-4, 5). Find the length, width, perimeter, and area of the rectangle. Show your work using absolute value.

Model answer: The width comes from the two x-values, |3 - (-4)| = 7, and the height comes from the two y-values, |5 - (-2)| = 7, so this rectangle is actually a square. That gives a perimeter of 2(7 + 7) = 28 units and an area of 7 × 7 = 49 square units.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.